



¿Can you create a
coding agent with
JavaScript in 30
minutes? Yes, you
can!



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I'm Asier

Software Engineer with chronic
addiction to side projects that I
never finish



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Who has ever **used AI** to
code?



Who has ever **used** any
coding agent?



Who would know how to
**define what is a coding
agent?**



How did we end up here?

What is a “Coding Agent”

How are we going to design our harness?



The whole revolution we
have been through the last
years com from the
**innovations in language
models**



In 2017 with the article
“Attention Is All You Need”
began the march towards
Large Language Models



We have **realised** that **we can**
do a lot with just text



In the coding world it all
started with the release of
GitHub Copilot in June of
2021





GitHub Copilot worked
with an **LLM** that center
only in ***autocomplete***





Lacked both **the context** of
the code you were writing
and the ability to **produce**
code of **more than a few**
lines





In **November of 2022**
ChatGPT released, the **first**
chatbot





Not long after, in **July of 2023 Github Copilot** evolved to having the **chatbot** format aside from the autocomplete





These chatbots where **much more intelligent**, they **completed harder** and **more complex** tasks





They remained tied to the
person that used them for
both **context** and
completion of those tasks





What would happen if we give those chatbots **autonomy** to complete tasks?



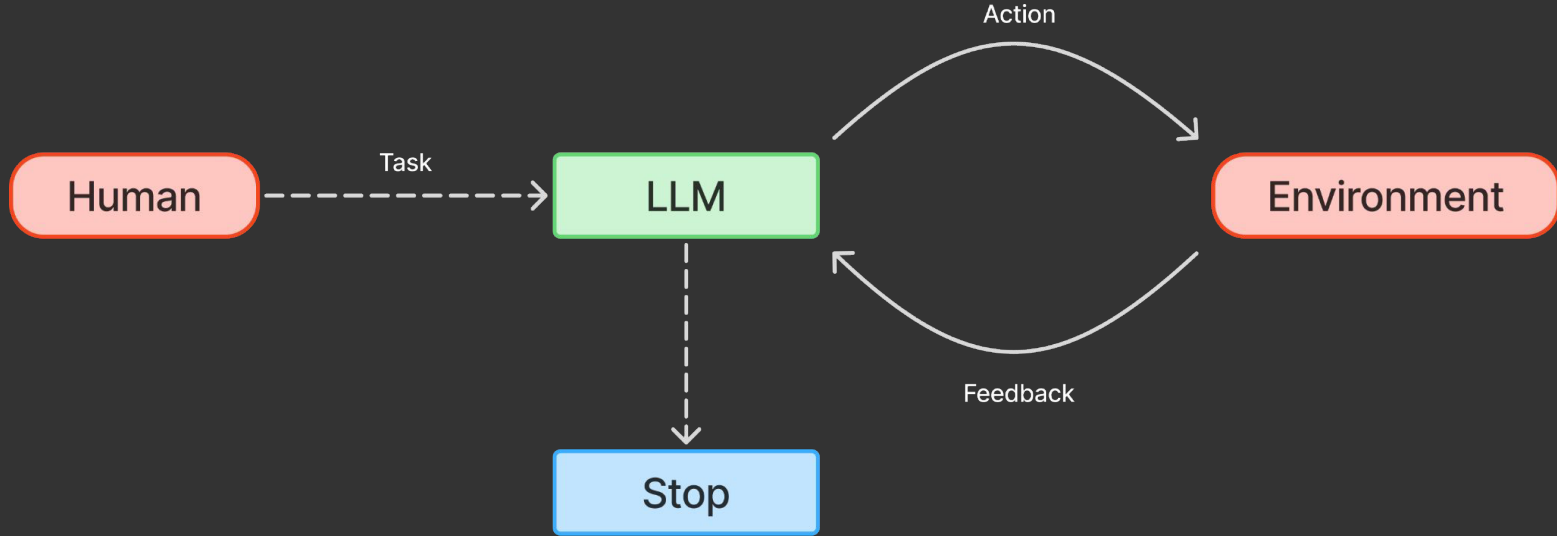
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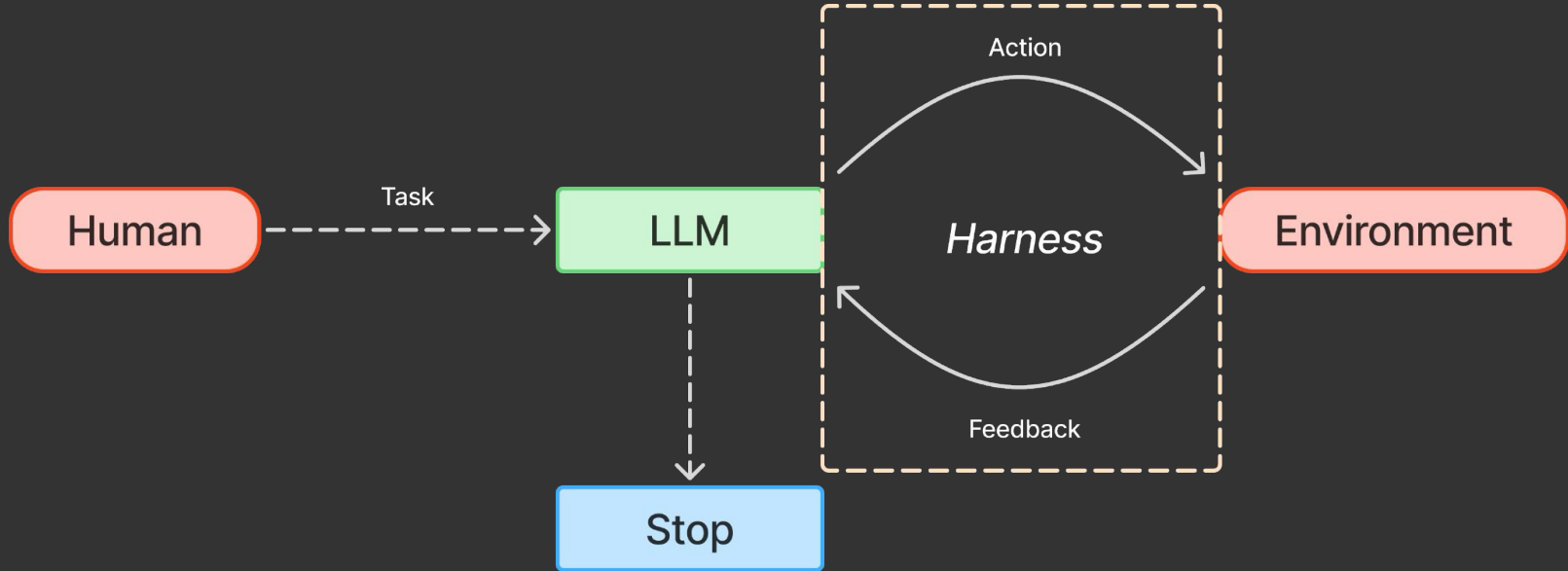


How do we give **autonomy to code** to these chatbots?





We give them a “***harness***”
with which it can interact with
the environment





Agent = LLM + Harness



What is a **Coding Agent**?



The “**harness**” has to
offer to the LLM a way of
Read, Write and **Search**
code



How did we end up here?
What is a “Coding Agent”

How are we going to design our harness?



A “harness” is composed of

Tools

Agent Loop

Context

Permissions/Sandbox



A “harness” can be as
complex or as **simple** as
we want



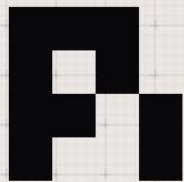
Thariq 
@trq212

Most people's mental model of Claude Code is that "it's just a TUI" but it should really be closer to "a small game engine".

For each frame our pipeline constructs a scene graph with React then

- > layouts elements
- > rasterizes them to a 2d screen
- > diffs that against the previous screen
- > finally uses the diff to generate ANSI sequences to draw

We have a ~16ms frame budget so we have roughly ~5ms to go from the React scene graph to ANSI written.



*There are many agent harnesses,
but this one is yours.*

Pi is a minimal terminal coding harness.

Adapt Pi to your workflows, not the other way around.



For the **MVP** we are going to **focus on**

Tools

Agent Loop

Context

Permissions/Sandbox



The widespread way of
“offering” to an **LLM** **what**
it can do is through ***Tools***



The ***Tools*** are just **code**
the the **LLM** can ask to
execute



We have to give it ***Tools***
to **Read, Write** and
Search



We are going to use a **local**
LLM



There are a lot of ways,
but the **easiest** is to just
use **Ollama**





The model we will use is
Gemma 4 26B from Google



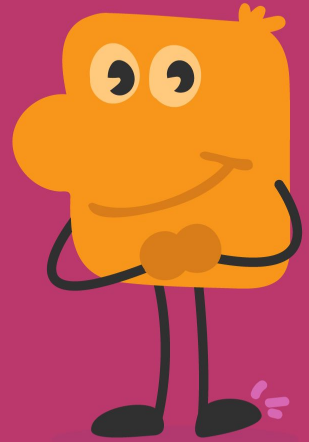
Gemma 4



For everything else we will use
Node.js



Let's write **Eddie** our
coding agent





Tools are prompt engineering



The **feedback loop between**
***Tools* matters** more than you
think



In the **System Prompt** is where
your **opinion/taste** lives

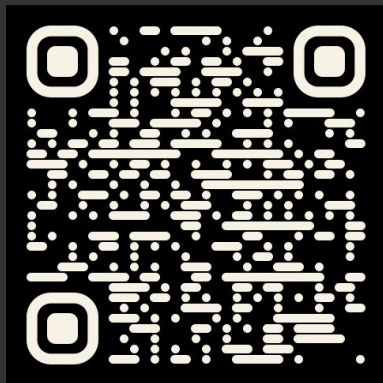


Things we miss in case you
want to continue:

***Sub-Agents, Permissions,
Persistent Memory,
Context Management***



Thank you



github.com/asierzapata/coding-agent-talk

